

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 100
timesteps_orig: 60
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.1
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 100
samples_per_sec: 10
forecast_horizon_sec: 3
forecast_steps: 30
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-05-10 16:55:40

Training Start: 2025-05-10 16-45-02

Training End: 2025-05-10 16-52-38

Training Duration: 7.60 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.10

MAE: 28799.8842

MSE: 1329038116.9693

RMSE: 36455.9751

R2 Score: 0.6856

Multi-Step Forecast:

Forecast Steps: 30

MAE: 0.0329

MSE: 0.0016

RMSE: 0.0403

R2 Score: 0.8270

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 10,265 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

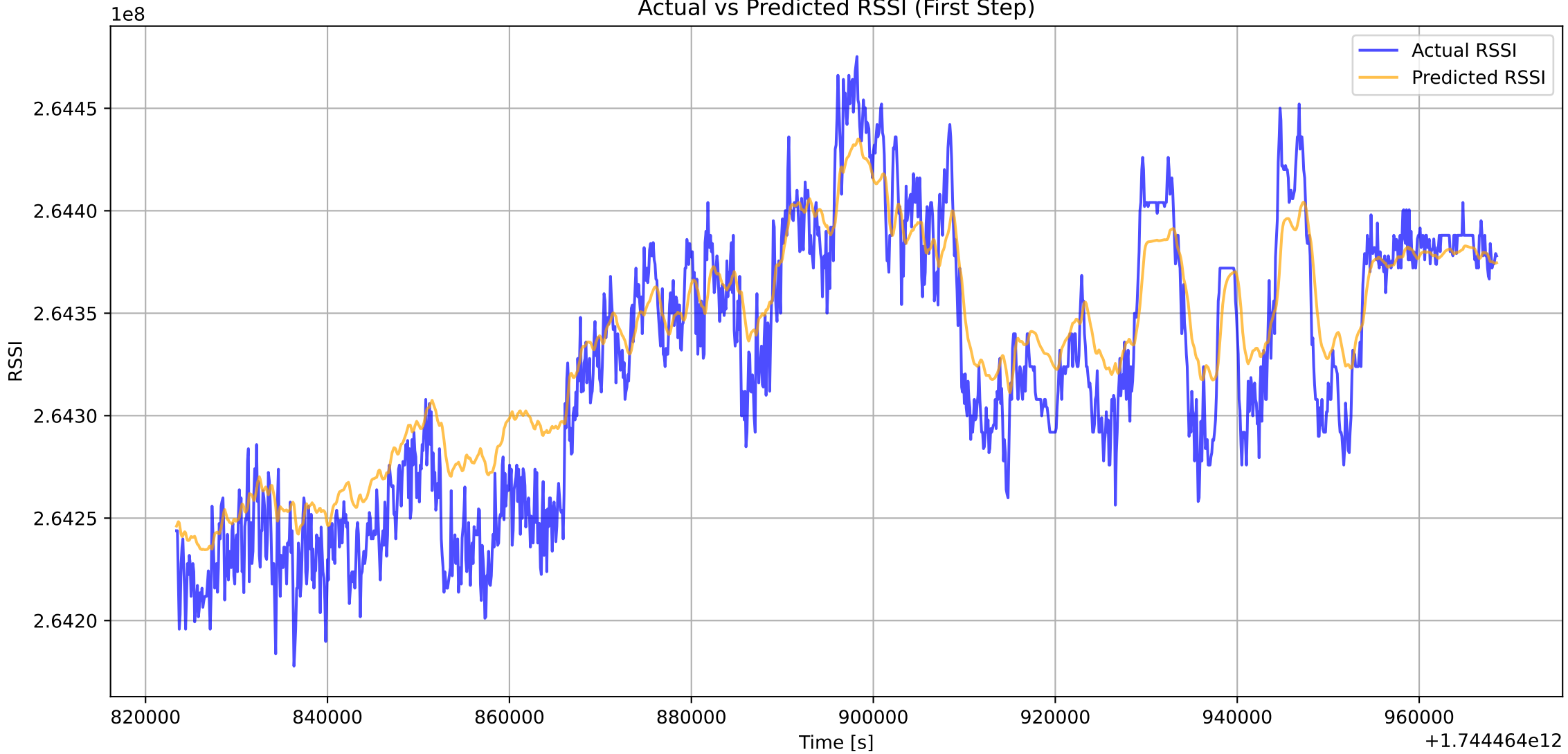
1744397085760
1744397085801
1744397086014
1744397086103
1744397086208

...

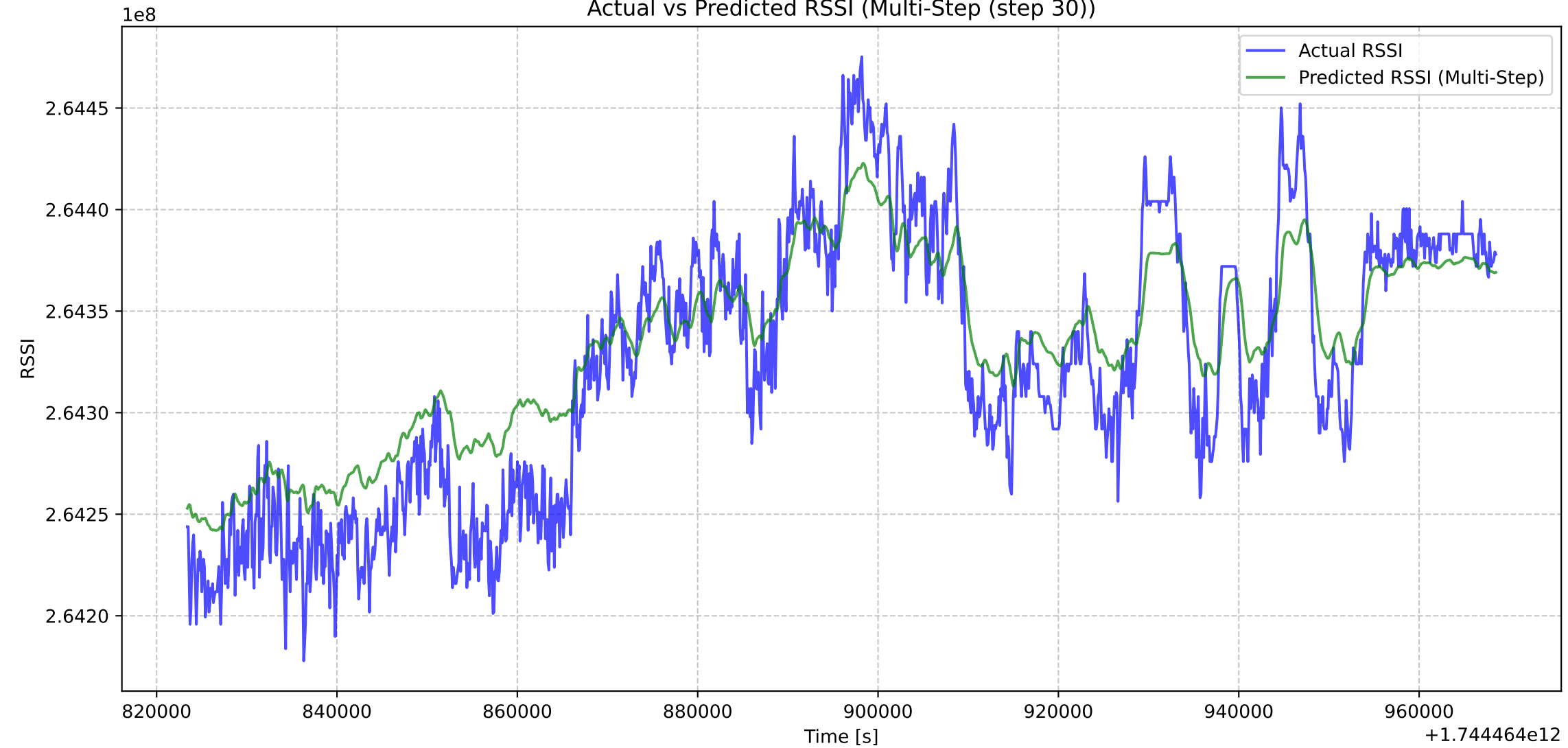
Last 5 timestamps AFTER:

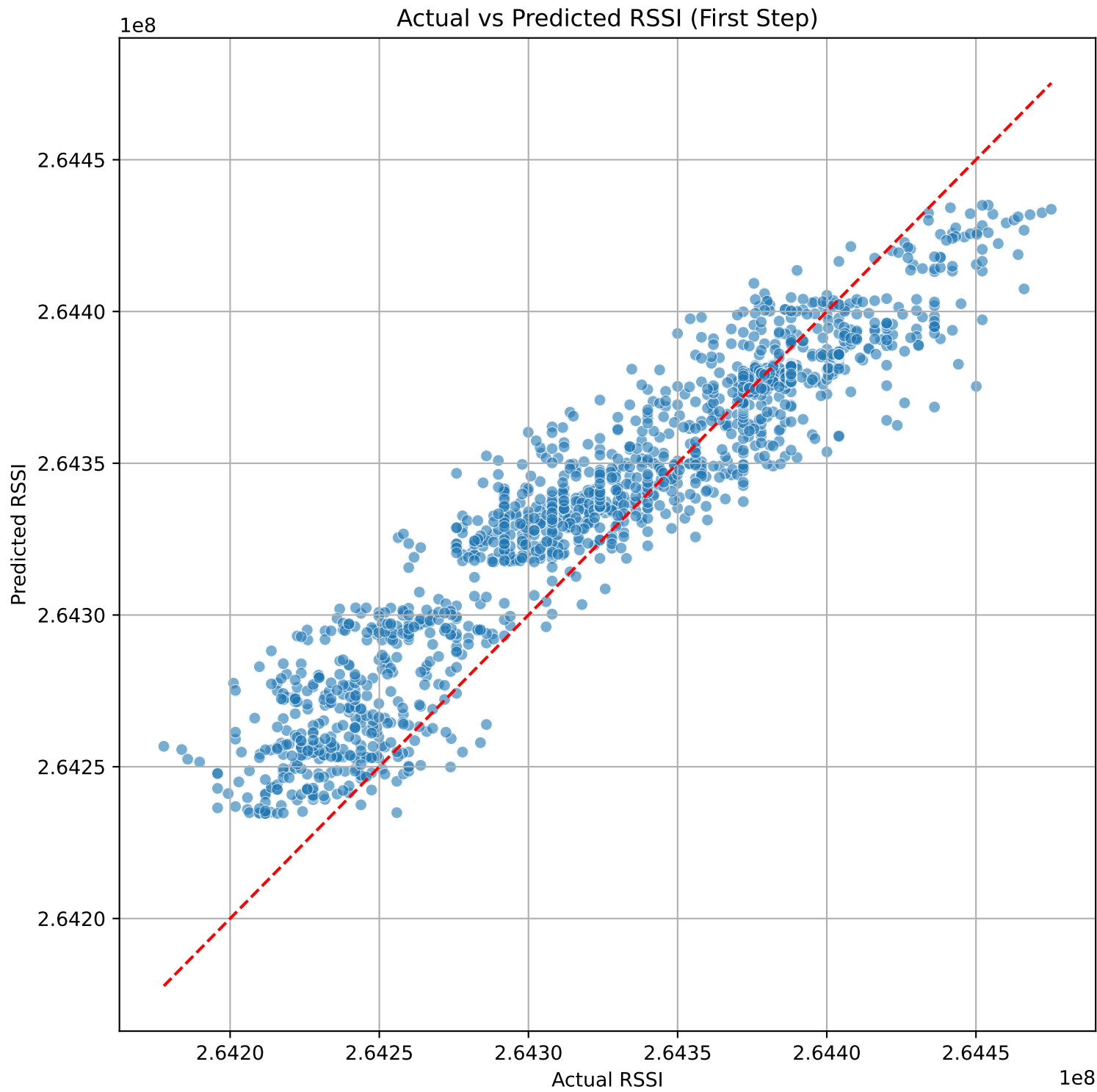
1744464971004
1744464971101
1744464971209
1744464971306
1744464971402

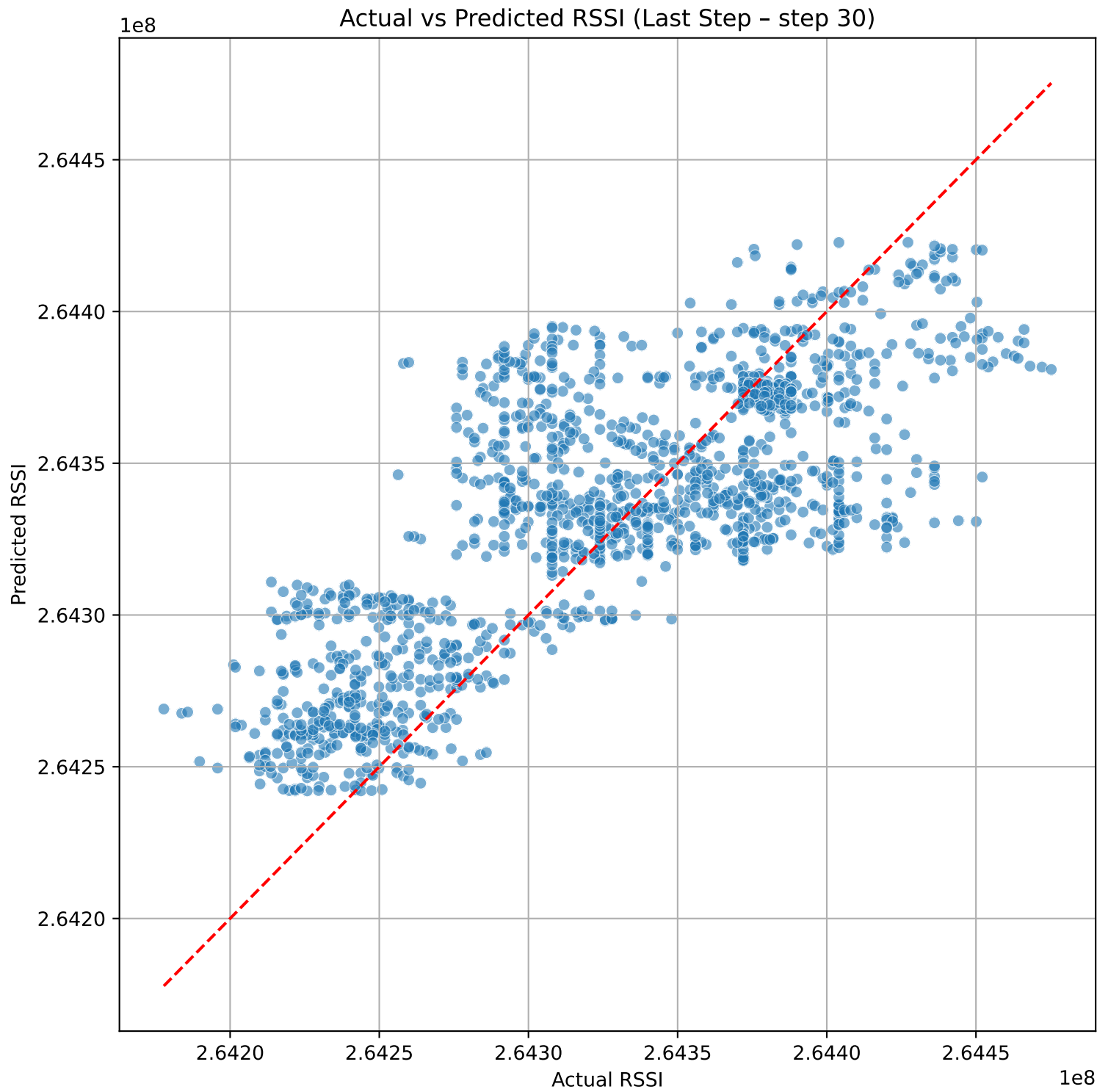
Actual vs Predicted RSSI (First Step)



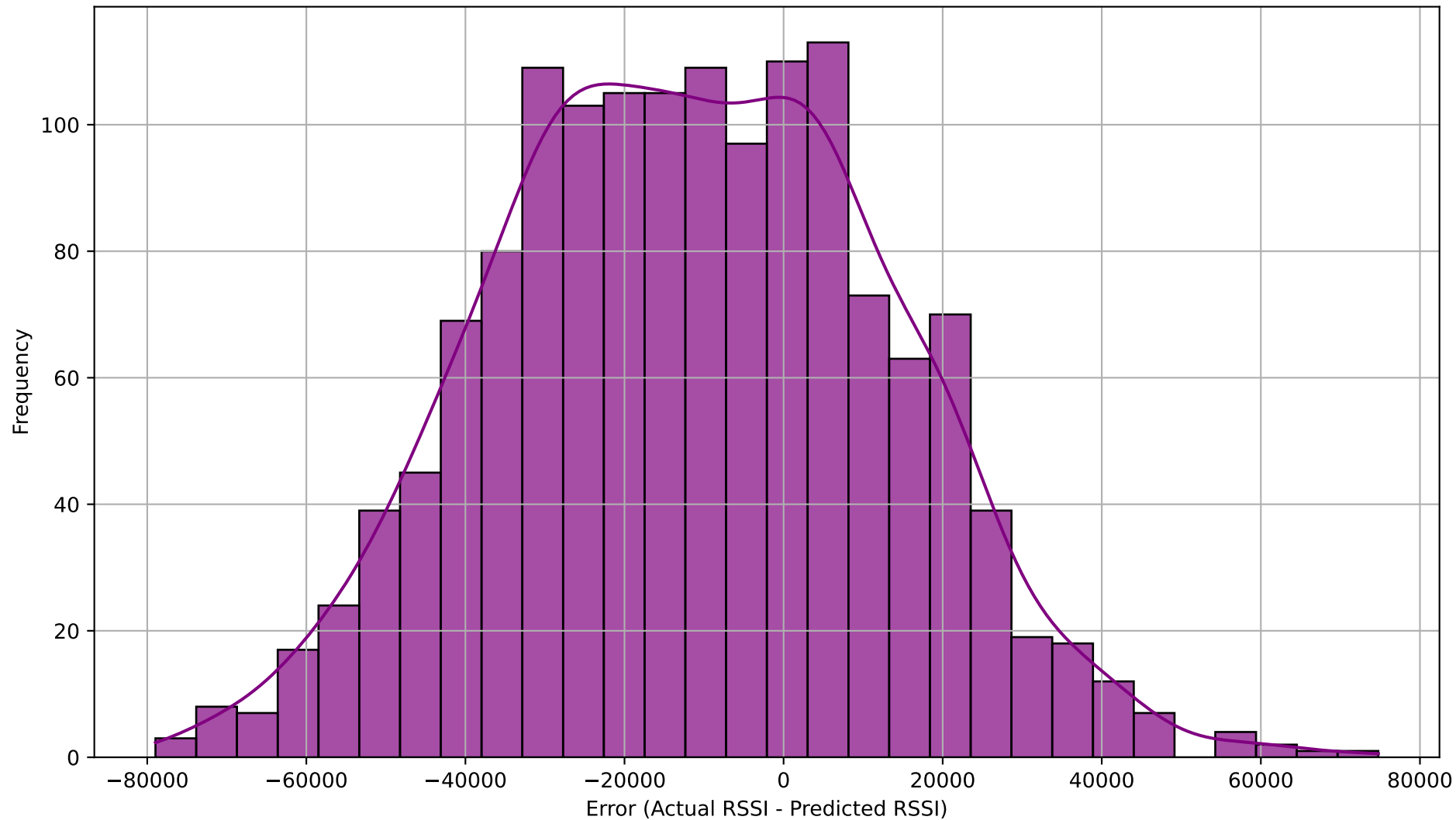
Actual vs Predicted RSSI (Multi-Step (step 30))



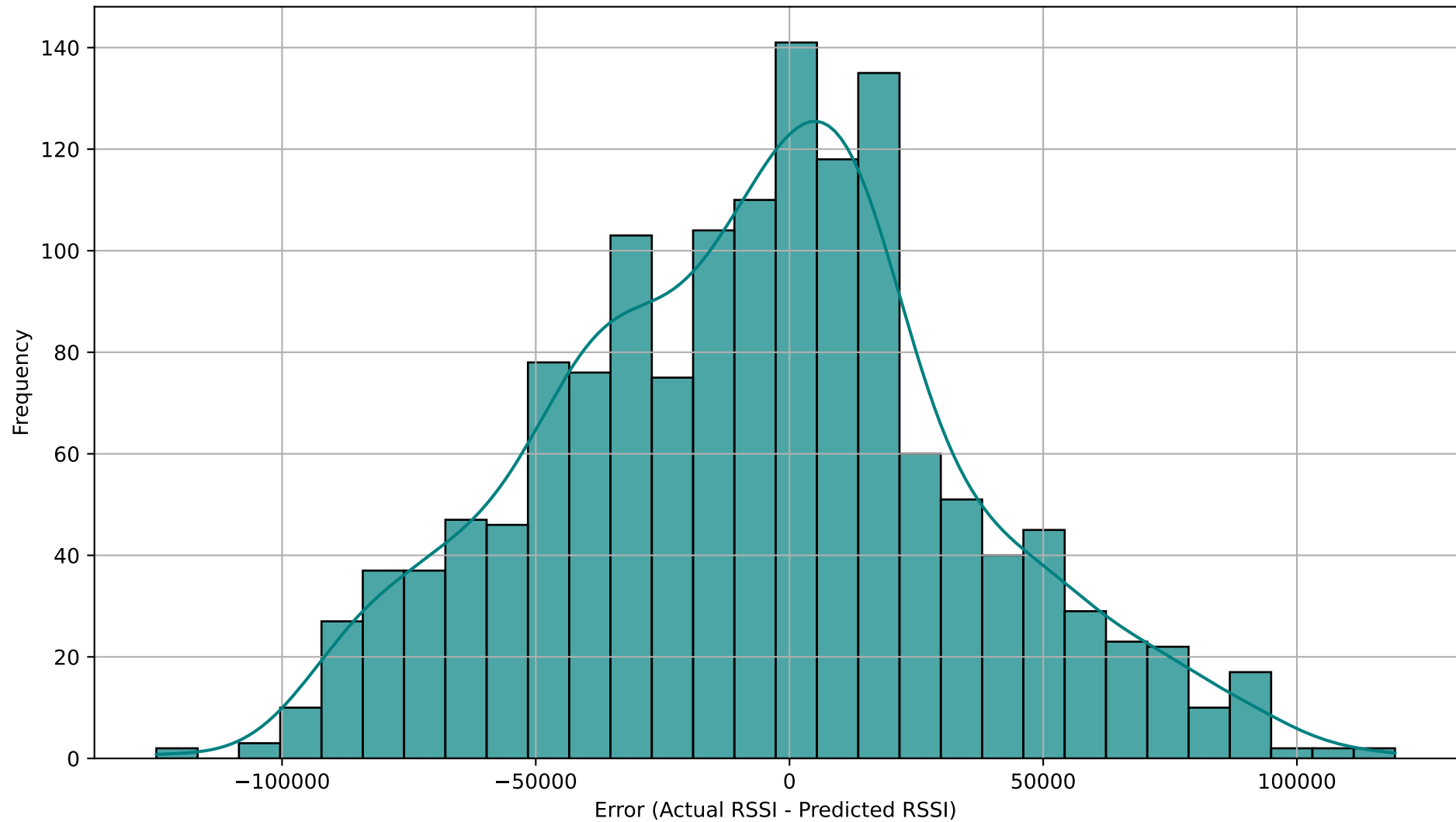




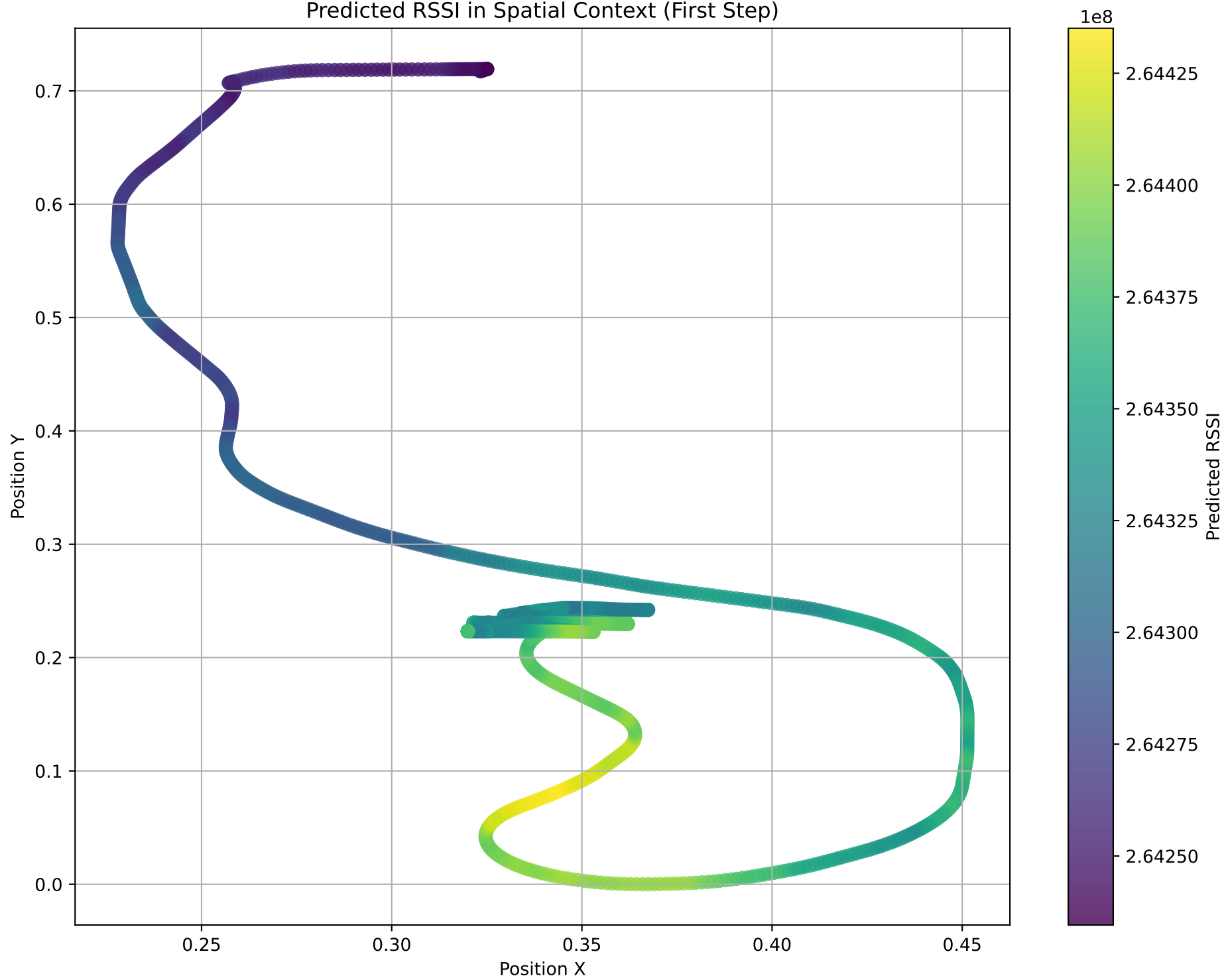
Distribution of Prediction Errors (First Step)



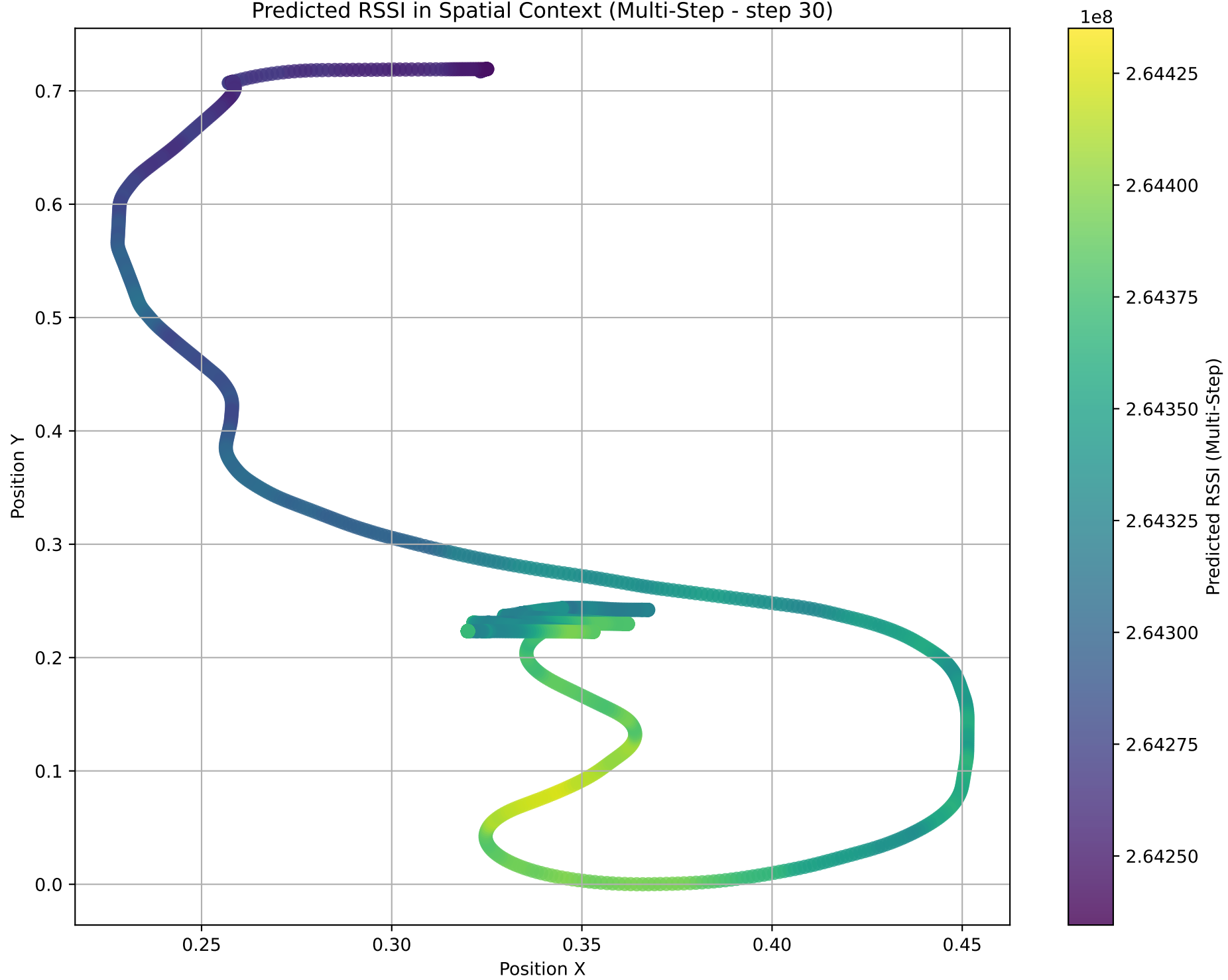
Distribution of Prediction Errors (Last Step - step 30)



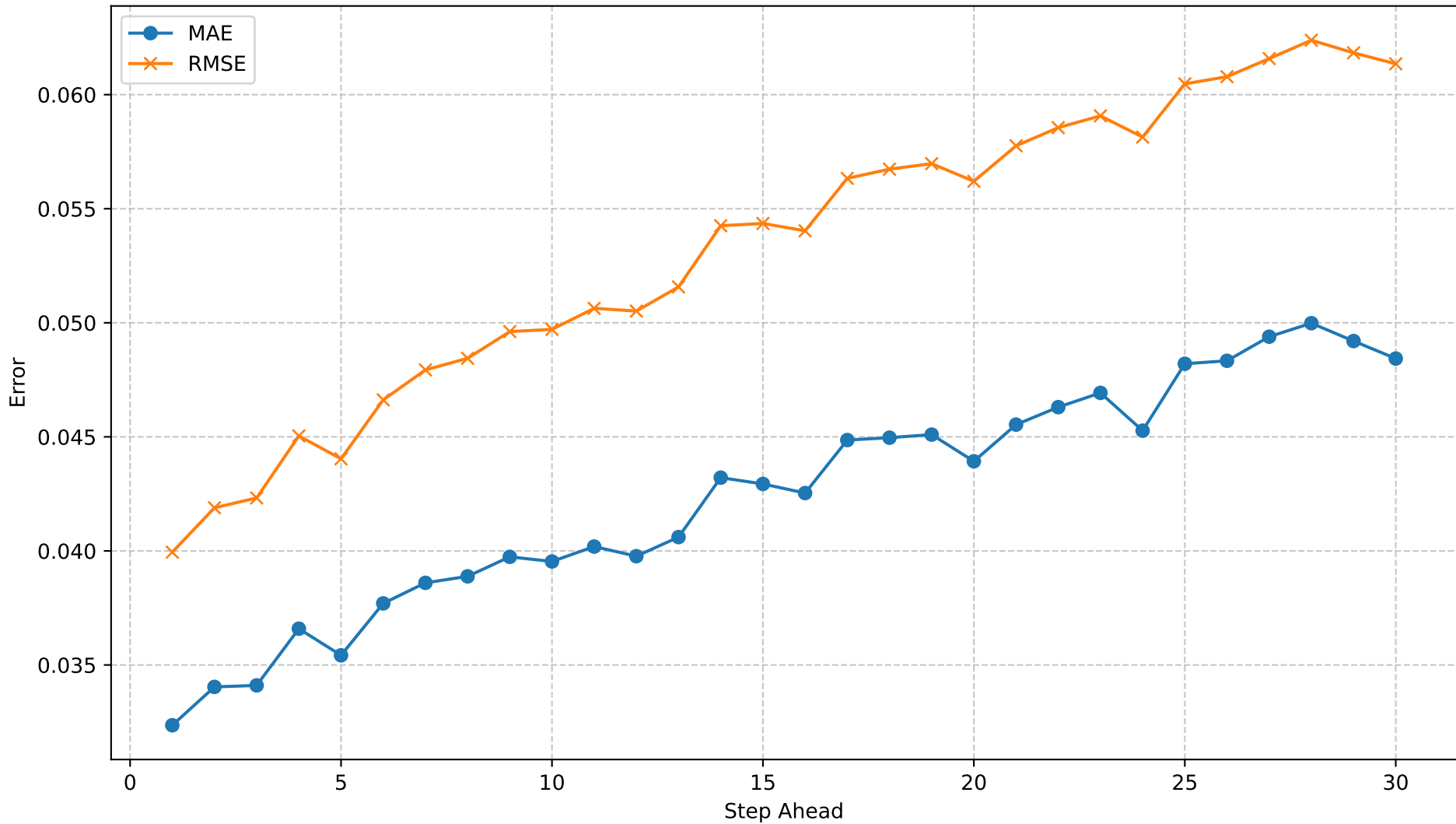
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 30)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 30

MAE / RMSE (dB)

- 1º passo : 0.03 / 0.04
- Mediana : 0.04 / 0.05
- 30º passo : 0.05 / 0.06

Crescimento do erro: 49.7% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 28 (0.05), 27 (0.05), 29 (0.05)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.